

ICE MACHINE PREVENTIVE MAINTENANCE PROCEDURES

VA Central California HCS
Fresno, CA 93703

SOP 138M-25-020

Signatory Authority:
Chief, Engineering Service

Service Line(s):
Engineering Service

Responsible Owner:
Chief, Maintenance & Operation Section

Rescinded Document(s):
SOP 138M-20-020

Effective Date:
October 28, 2025

Recertification Date:
October 31, 2030

1. PURPOSE AND AUTHORITY

a. The purpose of this standard operating procedure (SOP) is to provide the proper procedures for ice machine inspection and cleaning to ensure safe and reliable operation of ice machines and water dispensers used for patient care.

b. Joint Commission Standard EC.02.05.01.

2. PROCEDURES

a. Weekly Cleaning of Ice Machine Exteriors.

(1) Environmental Management Service (EMS) performs weekly exterior cleaning (SOP 137-XX-015 Ice Machine and Water Dispensing Unit Cleaning) in accordance with manufacturer recommendations.

b. Follett Ice Machine Preventive Maintenance.

(1) **Monthly.**

(a) Remove automatic machine ice chutes, water chutes, and spigot components monthly and replace with clean components.

(b) Clean dirty components by spraying with Nu-Calgon Liquid Ice Machine Cleaner and soaking for approximately 15 minutes to remove any scale buildup.

(c) After soaking, wipe down with a clean water damp cloth.

(d) Wipe down cleaned components with Nu-Calgon IMS-III Sanitizing Solution. Do not rinse the components.

(e) Allow components to air dry to prevent bacterial growth and store cleaned components in plastic wrap to ensure they remain sanitary. These cleaned parts can be used for the following month's preventative maintenance.

(f) When visibly worn, cracked, or otherwise damaged, ice chutes, water chutes, and spigot components shall be replaced with new components.

(g) Clean condenser coils (air-cooled only)

(2) **Quarterly.**

(a) Clean the internal system.

1. Ensure the evaporator is completely clear of ice. If not, complete steps 1-5 using only hot water before using the cleaning solution.

2. Remove the front cover and turn off the bin signal switch.

3. Dispense all ice and discard.

4. Remove splash panel.

5. Press the clean switch and wait for the machine to drain.

6. Fill reservoir with SafeCLEAN Plus solution and replace lid. The cleaning cycle will run and then rinse three times.

(b) Clean the Dispenser while the internal system is running the clean cycle.

1. Turn off the dispenser power.

2. Remove the hopper lid, knurled nuts, stud assembly, baffle, wheel, and any remaining ice.

3. Remove dispense chutes from the splash panel.

4. Submerge the drain grille in SafeCLEAN Plus solution and allow it to soak.

5. Wipe inside of hopper lid, stud assembly, baffle, wheel, inside of storage area, dispense chutes, drain grille, and drain pan with a damp cloth of SafeCLEAN Plus solution.

6. Thoroughly rinse all parts with a damp cloth wrung out in clean water.

7. When internal cleaning is complete, reinstall dispense chutes, wheel, baffle, stud assembly, and knurled nuts.

8. Remove the top bearing insulation and remove the nozzle connected to the evaporator only.

9. Soak SaniSponge in SafeCLEAN Plus solution and insert into the nozzle, then insert a dry sponge into the nozzle.

10. Replace nozzle on evaporator, ensuring drain is connected to reservoir and vent tubes are connected to evaporator drain pan.

11. Replace top bearing insulation.

12. Turn on the bin signal switch and wait for ice to push sponges through the transport tube.

13. Collect sponges from the storage bin.

14. Replace the hopper lid, turn on dispenser power, and install the front cover.

15. After 10 minutes, dispense all ice and discard.

c. Hoshizaki Ice Machine Preventive Maintenance.

(1) Monthly.

(a) Check the proper pressure of external water filters and change if necessary.

(2) Quarterly

(a) Clean & sanitize ice maker and storage bin

(b) Wipe down gear motor drain pan

(c) Inspect the icemaker, storage bin, and drip tray drain to ensure they are clear.

(d) Inspect storage bin/extruding seal bolts for leakage, tighten or replace as necessary. Seal bolts must be replaced if removed.

(3) Annually.

(a) Close the water supply line and drain the water system. Clean the inlet water valve and dispense water valve screens. Clean and inspect the drain valve.

(b) Inspect water hoses, clean/replace if necessary.

(c) Inspect/clean condenser coils.

(d) Inspect for oil spots, loose components, fasteners, and wires.

(e) Check for upper bearing wear using .02" round stock or pin gauge. Replace both upper and lower bearings if wear exceeds factory recommendations.

(f) Inspect O-ring (lower housing), mechanical seal, evaporator cylinder, and auger for wear or damage and replace if necessary.

NOTE: *In accordance with Appendix C, Facility Legionella Hazard Assessment-Section 4.5 Ice Machines Inspection, Sanitizing, and component replacement shall be completed quarterly.*

3. ASSIGNMENT OF RESPONSIBILITIES

a. **Facilities Management Section, Refrigeration and Air Conditioning Shop.** Will be responsible for assuring that all patient care-related automatic ice machines are cleaned and sanitized per manufacturer's recommendations and the procedures set forth in Section 4.5 of the Facility Legionella Hazard Assessment.

4. DEFINITIONS

a. None

5. REFERENCES

- a. Hoshizaki Ice Machine Cleaning and Maintenance Manual
- b. Follett Ice Machine Cleaning and Maintenance Manual
- c. Facility Legionella Hazard Assessment-Section 4.5 Ice Machines
- d. SOP 137-25-015 Ice Machine and Water Dispensing Unit Cleaning
- e. Joint Commission Standard EC.02.05.01

6. REVIEW

Frequency of review is, at a minimum, required during recertification or when there are changes to the governing document and any regulatory requirement for more frequent review.

7. RECERTIFICATION

This SOP is scheduled for recertification on or before the last working day of October 2030 – 5 years from the effective date. In the event of contradiction with national policy, the national policy supersedes and controls.

8. SIGNATORY AUTHORITY

Travis A. Horn
Chief, Engineering Service
Date Approved: Embedded with /es/

NOTE: *The signature remains valid until rescinded by an appropriate administrative action.*

DISTRIBUTION: SOPs are available at:

<https://dvagov.sharepoint.com/sites/VHAFREQuality/MCP/SOPNEW/Forms/AllItems.aspx>

APPENDIX A

Hoshizaki Ice Machine Maintenance

The maintenance schedule below is a guideline. More frequent maintenance may be required depending on water quality, the appliance's environment, and local sanitation regulations.

WARNING	
• Only qualified service technicians should service the appliance. • To reduce the risk of electric shock, do not touch the control switch or service switch with damp hands. • Before servicing: Move the power switch to the "OFF" position and unplug the appliance to prevent electric shock or injury from moving parts.	

Maintenance Schedule		
Frequency	Area	Task
Monthly	External Water Filters	Check proper pressure and change if necessary.
	Appliance Exterior	Wipe down with a clean, soft cloth. Use a damp cloth containing a neutral cleaner to wipe off oil or dirt buildup. Clean any chlorine staining (rust-colored spots) using a non-abrasive cleaner.
Every 6 Months	Icemaker and Ice Storage Bin	Clean and sanitize per the cleaning and sanitizing instructions provided in the instruction manual or maintenance label on the appliance.
	Gear Motor Drain Pan	Wipe down with a clean cloth and warm water.
	Icemaker, Storage Bin, and Drip Tray Drain	Check to make sure they are clear.
	Storage Bin/ Extruding Head Seal Bolts	Inspect for leakage around seal bolts. Tighten (see torque values below) or replace as necessary. Seal bolts must be replaced once removed because the seal material is single-

		use only. If new seal bolts do not have pre-applied thread locker, apply Loctite 243 or equivalent thread locker to seal bolt threads. • Torque: 11.1 ft-lb/15 N-m
Yearly	Inlet Water Valve, Dispense Water Valve, and Drain Valve	Close the water supply line shut-off valve and drain the water system. Clean the inlet water valve and dispense water valve screens, and clean and inspect the drain valve.
	Water Hoses	Inspect the water hoses and clean/replace if necessary.
	Condenser	Inspect. Clean, if necessary, by using a brush or vacuum cleaner.

	Icemaker	Inspect for oil spots, loose components, fasteners, and wires.
	Upper Bearing (extruding head)	Check for wear using .02" round stock or pin gauge. Replace both upper bearing and lower bearing if wear exceeds factory recommendations. See "IV.C.1.Upper Bearing Wear Check".
After 3 Years, then yearly	Upper Bearing (extruding head); Lower Bearing and O-Ring (lower housing); Mechanical Seal; Evaporator Cylinder; Auger	

Follett Ice Machine Maintenance

Follett ice machines and dispensers, and their associated cleaning and sanitizing procedures, are designed for use with potable water sources. The presence, or suspected presence, of infectious agents may call for additional measures, including the replacement of components and more comprehensive disinfection measures. Follett recommends that these cleaning and sanitizing procedures be reviewed with the appropriate infectious agent subject matter experts to ensure complete remediation.

Periodic cleaning of Follett's ice and water dispenser and ice machine system is required to ensure peak performance and delivery of clean, sanitary ice. The recommended cleaning procedures that follow should be performed at least as frequently as recommended and more often if environmental conditions dictate.

Follett recommends sanitizing the pressurized water lines before cleaning the ice machine/dispenser. Follett offers two kits: order P/N 01089572 when a Follett filter system with a pre-filter bowl is present, or P/N 01089580 when a Follett filter system is not present. Follow the instructions provided with the respective kits to sanitize the pressurized water lines immediately before cleaning the ice machine/dispenser.

Cleaning of the condenser can usually be performed by facility personnel. Cleaning of the ice machine system should be performed by your facility's trained maintenance staff or a Follett authorized service agent. Regardless of who performs the cleaning, it is the operator's responsibility to see that this cleaning is performed according to the schedule below. Service problems resulting from a lack of preventive maintenance will not be covered under the Follett warranty.

Recommended Cleaning Intervals

Symphony Plus	Frequency
Drain Line	Weekly
Drain Pan/Drip Pan	Weekly
Exterior, Water Station Tube	As needed
Condenser	Monthly (air-cooled only)
Dispenser and Components	Semi-Annually
Ice Machine	Semi-Annually
Transport Tube	Semi-Annually
Ice Storage Area/Bin	Semi-Annually

Pressurized Water Sanitizing	Semi-Annually
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Weekly

CAUTION
Do not use solvents, abrasive cleaners, metal scrapers, or sharp objects to clean any part of the dispenser.

Dispenser, drain pan, and drain line

- Pour 1 gal. (3.8 L) of hot tap water into the drain pan to flush the drains.

Splash panel front, Sensor SAFE infrared dispensing

1. Deactivate dispensing by pressing and releasing the clean switch located on the left side of the unit under the top front cover.

2. Clean the lens and splash panel front using a soft cloth and mild, non-abrasive, non-chlorine-based cleaner.

3. Reactivate dispensing by pressing and releasing the clean switch again.

Monthly

CAUTION
• Do not use solvents, abrasive cleaners, metal scrapers, or sharp objects to clean any part of the dispenser.

Condenser (air-cooled ice machine only)

- Use a vacuum cleaner or stiff brush to carefully clean condenser coils of lint and debris to ensure optimal performance.

Semi-Annually (more often if conditions dictate)

- A cleaning procedure should always include both the ice machine and dispenser.
- The ice-making system can be cleaned in place.

CAUTION
• Wear rubber gloves and safety goggles (or face shield) when handling SafeCLEAN Plus and IMS-III solutions. • Use only Follett-approved cleaners. • Do not use solvents, abrasive cleaners, metal scrapers, or sharp objects to clean any part of the dispenser

Cleaning Tool Checklist

- (1 or 2) 1.5-gallon (or larger) plastic bucket
- (2) clean cloths
- Sanitary gloves
- Safety glasses
- (2) SaniSponge (P/N 00131524 – single sponge)
- SafeCLEAN Plus ice machine cleaner

SafeCLEAN Plus Solution: Follow the directions on the SafeCLEAN Plus packaging to mix 1 gal. (3.8 L) of Follett SafeCLEAN Plus solution. Use 100°F (38°C) water.

Ice Machine and Dispenser Cleaning Procedure

Note: Check drains and the drain cup to ensure they are open and flowing freely.

1. If the ice machine was running recently, ensure that the evaporator is completely free of ice before proceeding. If there is ice in the evaporator, complete steps 2-7 using only hot water to remove the ice, then begin Cleaning Procedure again.
2. Remove the front cover and turn **OFF** the bin signal switch.
3. Dispense all ice from the storage hopper and discard.
4. Remove splash panel.
5. Press the **CLEAN** switch. The **MAINTENANCE** light will turn on, and the machine will drain. Wait for the **LOW WATER** light to turn on.
6. Remove the lid from the cleaning cup and fill (about 1 quart) until the SafeCLEAN Plus solution completely fills the reservoir. Place the lid back on the cup. Save the remainder of the SafeCLEAN Plus solution.
7. **CLEANER FULL** light will turn on and the machine will start the cleaning cycle, then rinse three times; this
The process takes approximately 15 minutes.
8. While the ice machine is cleaning, clean the dispenser as follows:
 - a. Turn **OFF** dispenser power.
 - b. Remove hopper lid.
 - c. Remove knurled nuts from the front of the storage hopper.
 - d. Remove stud assembly, baffle, wheel, and any remaining ice.
 - e. Remove dispense chutes from the splash panel.
 - f. Submerge the drain grille in SafeCLEAN Plus solution and allow it to soak to remove any scale buildup.
 - g. Wipe inside of hopper lid, stud assembly, baffle, wheel, inside of storage area, dispense chutes, drain grille, and drain pan with a damp cloth wrung out in SafeCLEAN Plus solution. Thoroughly rinse all parts with a damp cloth wrung out with clean water.
- Note:** To avoid possible damage to the motor assembly, only use a damp cloth to clean the storage hopper. Do not allow water to run through the motor shaft hole in the bottom of the hopper.
9. When the machine is finished cleaning, the **MAINTENANCE** light will turn off.
10. Reinstall dispense chutes, wheel, baffle, stud assembly, and knurled nuts.
11. Remove top bearing insulation. Loosen the Phillips-head screw on the nozzle connected to the evaporator. Remove nozzle from evaporator side only, leave other side of nozzle connected to transport tube.
12. Soak one SaniSponge in the remaining SafeCLEAN Plus solution.
13. Insert the sponge soaked in cleaning solution into the nozzle, then insert a dry sponge into the nozzle.

14. Replace the nozzle onto the evaporator and tighten the screw. Ensure the drain is connected to the reservoir and the vent tubes are connected to the evaporator drain pan. Replace top bearing insulation.
15. Turn **ON the** bin signal switch. Wait for the ice to push the sponges through the transport tube.
16. Collect sponges from the ice storage bin.
17. Replace hopper lid, machine top, turn **ON** dispenser power, and install front cover.
18. After 10 minutes, dispense all ice and discard.

User Interface and Exterior Cabinet

- Clean stainless-steel panels with stainless steel cleaner.

Section I - Hazard Assessment

4.5 Ice Machines

SITE USES

- Throughout the Facility

HAZARD ANALYSIS

Biological Risk	Source Exposure Risk	Human Exposure Risk	CCP
Yes Ice machines should be maintained to mechanical and operational standards. Ice and equipment can become contaminated by non-Tuberculosis Mycobacteria (NTM), Pseudomonas, Enterobacter, Legionella, and Cryptosporidium.	No The machine itself does not produce aerosols. Risk is possible if ice is aspirated when eaten.	Yes Risk is possible if ice is aspirated when eaten.	Yes

HAZARD CONTROL STRATEGIES

Strategy	Monitoring Goal	Frequency	Current Practice	Action Needed
Clean & Disinfect	Scheduled maintenance. Follow the manufacturer or general guidelines below	Quarterly for High-risk patient areas Semi-annual for all patient areas	Verify current practice	
Point-Of-Use (POU) Filters	Scheduled maintenance. Use 0.2-micron absolute filters on high-risk patients	Per manufacturer specifications; validated based on the standard ASTM	Not used	

	area ice machines	838-05 for 0.2 micron sterilizing filters		
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GENERAL INDUSTRY PRACTICE

The following practices are recommended when handling ice:

- ☐ Wash your hands before obtaining ice
- ☐ Avoid hand contact with ice
- ☐ Use smooth-surface ice scoops and stow on a sanitary surface or location.

Attached the scoop to the ice machine with a chain.

- ☐ Limit access to authorized personnel
- ☐ Keep ice machine doors closed between uses
- ☐ When purchasing new ice machines, consider machines with automatic dispensers into portable containers

The following procedures are recommended for cleaning and disinfection.

Waterlines that feed the ice machines should be inspected semi-annually and replaced if found to be fouled. If the ice machines become fouled frequently, then the frequency of cleanings and inspections should be increased. Ice machines and storage chests should also be cleaned and disinfected per Manufacturer guidelines. All procedures, cleanings, and maintenance should be documented. Use sterile or sterile-grade filtered water with 0.2-micron filtration for all water used to clean and disinfect ice machine components. Use a sterile or sterile disposable basin for soaking components; do not use a sink due to drain contamination. Avoid contamination of ice machine drains and trays, which increases the risk of retrograde contamination on the ice chute and water dispenser.

General Steps for cleaning and maintaining ice machines, dispensers, and storage chests:

1. Disconnect the unit from the power supply.
2. Remove and discard ice from the bin or storage chest.
3. Allow the unit to warm to room temperature.
4. Disassemble the removable parts of the machine that make contact with water to make ice.
5. Thoroughly clean the machine and parts with water and detergent.
6. Dry external surfaces of removable parts before reassembling.
7. Check for any needed repair.
8. Replace feeder lines, as appropriate (e.g., when damaged, old, or difficult to clean).
9. Ensure the presence of an air space in the tubing leading from the water inlet into the water distribution system of the machine.
10. Inspect for rodent or insect infestations under the unit and treat, as needed.
11. Check door gaskets (open compartment models) for evidence of leakage or dripping into the storage chest.
12. Clean the ice-storage chest or bin with fresh, sterilizing-grade filtered (0.2 µm) tap water. Add detergent; rinse with sterilizing-grade filtered (0.2 µm) water.
13. Disinfect machine by circulating a 50–100 parts per million (ppm) solution of sodium hypochlorite (i.e., 4–8 mL sodium hypochlorite/gallon of water) through the ice-making and storage systems for 2 hours (100 ppm solution), or 4 hours (50 ppm solution).
14. Drain sodium hypochlorite solutions and flush with sterilizing-grade filtered (0.2 µm) water.
15. Allow all surfaces of equipment to dry before returning to service.

This recommendation is drafted from the “Guidelines for Environmental Infection Control in Health-Care Facilities-Recommendations of CDC and the Healthcare

Infection Control Practices Advisory Committee (HICPAC) U.S. Department of Health and Human Services Centers for Disease Control and Prevention (CDC) Atlanta, GA 30333 2003” pages 66 and 67, with modifications for sterile rinsing from Pall Medical.